



PRD - Uber Safety Assurance Mode-Guardian Mode

Problem Statement:

How might we design a feature that enhances rider safety and trust during trips—especially in high-risk scenarios—while ensuring that the core ride-hailing experience remains seamless and user-friendly?

1.Context

As a Product Manager at Uber, user research highlights that safety concerns remain a significant barrier to adoption, particularly for female riders and individuals traveling late at night. Riders are seeking more transparent and proactive safety features that provide peace of mind during their journeys.

1.1 Current Scenario

Uber currently offers several safety features including real-time GPS tracking, driver identification, emergency assistance button, and trip sharing capabilities. These features are accessible through the main app interface and have been well-received by users. However, user research and support ticket analysis reveal gaps in the current safety ecosystem.

1.2 Limitations or challenges

- **Passive Safety Features:** Most existing safety tools are reactive rather than proactive. Users must actively remember to share their trip or initiate emergency protocols when they feel unsafe.
- **Limited Context Awareness:** The app doesn't currently assess risk factors such as time of day, route deviation, unusual stops, or driver behavior patterns that could indicate potential safety concerns.
- **Fragmented Safety Controls:** Safety features are scattered across different menus and screens, making them difficult to access quickly during moments of distress or concern.
- **Insufficient Real-Time Communication:** While riders can contact emergency services, there's limited capability for discreet communication or automated alerts to trusted contacts when riders feel uncomfortable but aren't facing immediate danger.
- **Low Feature Adoption:** Data shows that only 23% of riders regularly use trip sharing, and the emergency button is often discovered only after an incident occurs. This suggests poor feature discoverability and integration into the user journey.
- **Trust Gap:** Post-trip surveys indicate that 34% of female riders and 28% of riders using the service between 10 PM and 5 AM report feeling anxious during their trip, despite existing safety measures.

Competitive analysis shows that rival platforms like Lyft and regional players are investing heavily in AI-powered safety features, autonomous monitoring systems, and integrated emergency response capabilities. Without innovation in this space, Uber risks losing market share among safety-conscious user segments.

2 Objective/Goals

Build an **always-on, proactive, context-aware safety layer** that operates silently in the background and intervenes intelligently when risk signals are detected.

2.1 Business Objectives

- Increase rider retention by 5–8% among safety-conscious segments
- Reduce safety-related support tickets by 15%
- Increase trip-sharing activation from 23% → 50% via automation
- Improve brand perception as “Safest Ride Platform”
- Increase night & solo ride bookings by 6%

2.2 User Objectives

- Feel continuously monitored without effort
- Receive proactive check-ins before escalation
- Access safety tools instantly from main ride screen
- Share ride automatically without manual action
- Trigger silent help discreetly

3. User Persona

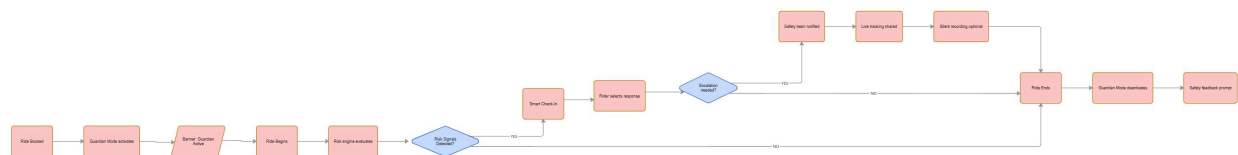
Persona	Persona-1	Persona-2	Persona-3
Background	Ananya , 27, IT professional working late shifts in a metro city. Frequently books rides alone after office hours. Tech-savvy but safety-conscious.	Raj , 21, college student who travels between campus and home. Parents are concerned about his safety during long intercity rides. Moderate app user.	Maria , 42, corporate manager and frequent business traveler. Uses Uber regularly in multiple cities. Values efficiency and reliability.
Needs	Wants proactive monitoring during rides, especially when routes change unexpectedly. Needs quick and discreet access to safety support.	Wants automatic ride sharing with parents and reassurance during long trips. Needs visible confirmation that someone is monitoring the ride.	Wants seamless safety integration without extra steps. Needs centralized, easy-to-access safety tools during every ride.

Frustrations	Feels anxious when drivers take unfamiliar routes. Finds safety features hidden and reactive. Does not want to manually activate safety tools every time.	Parents constantly call during rides for updates. Has to manually share trip details. Finds safety settings scattered across menus.	Finds existing safety features passive. Does not want interruptions unless truly necessary. Frustrated by lack of contextual alerts.
Goals	Complete rides with peace of mind and feel protected without extra effort.	Travel independently while keeping family reassured automatically.	Travel efficiently while knowing safety monitoring is always active in the background.

4. User Stories

1. As a rider, I want my ride to be automatically monitored so I don't need to activate safety manually.
2. As a rider, I want to be alerted if something unusual happens.
3. As a rider, I want to discreetly alert Uber if I feel unsafe.
4. As a rider, I want my ride auto-shared during higher-risk scenarios.
5. As a rider, I want visible reassurance that safety monitoring is active.

5. User Flow



6. Requirements

6.1 Technical Requirements

Risk Scoring Engine (MVP)

Inputs:

- GPS route deviation (>500m)
- Stop duration anomaly (>3 minutes in non-traffic zone)
- Speed anomaly (harsh braking, over speeding)
- Trip duration variance (>20% expected time)
- Geo-risk heatmap

Processing:

- Real-time risk scoring (0–100)
- Threshold-based escalation

Infrastructure:

- Real-time event streaming
- Secure cloud storage for safety logs
- Audio recording storage (encrypted)

6.2 Design Requirements

1. Visible “Guardian Active” banner on ride screen
2. Single-tap access to safety panel
3. Minimal cognitive load
4. Discreet activation gestures (long press / power button trigger)
5. Calm, non-alarming UI language
6. Accessible for visually impaired users

6.3 Functional Requirements

Core Features (MVP)

1. Always-on risk monitoring
2. Smart proactive check-in

3. Automatic ride sharing during risk escalation
 4. Silent alert gesture
 5. Centralized safety dashboard
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Phase 2 Enhancements

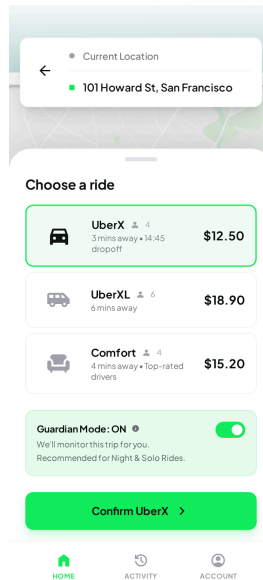
1. ML-based driver behavioral risk scoring
2. Personalized rider risk sensitivity
3. Predictive anomaly detection
4. Crime heatmap integration
5. Voice stress detection (future scope)

6.4 Non-functional Requirements

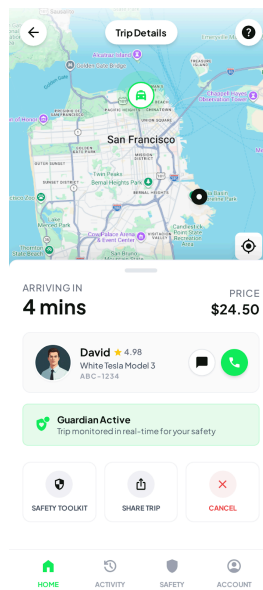
- Latency < 2 seconds for anomaly detection
- 99.9% uptime for safety services
- End-to-end encryption for audio/location
- GDPR and regional privacy compliance
- Battery usage impact < 5% per ride
- No degradation of booking performance

7. Wireframe & Prototype

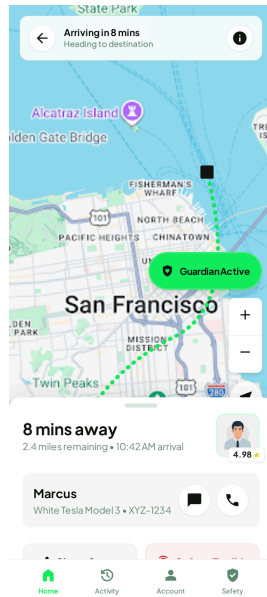
1.Ride Booking :



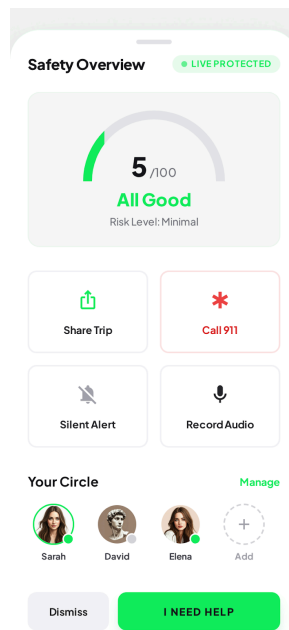
2. Ride Confirmed :



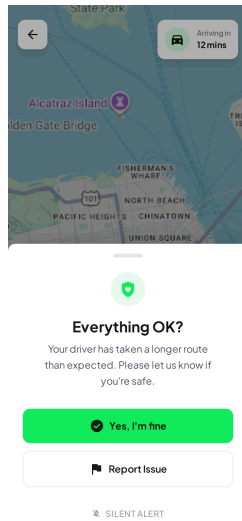
3.Active Ride :



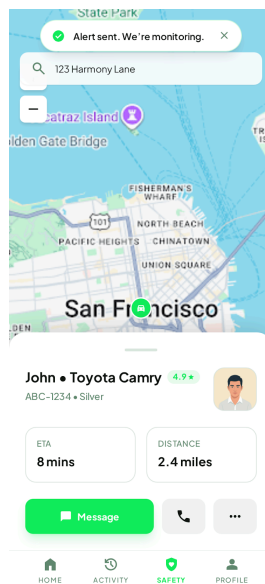
4. Safety dashboard:



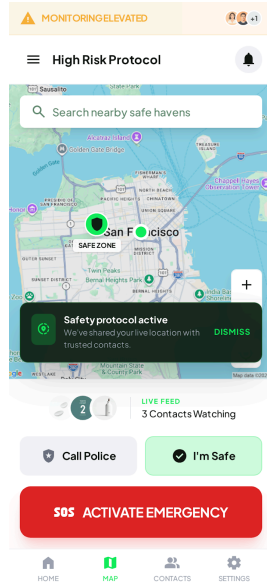
5.Proactive_checkin:



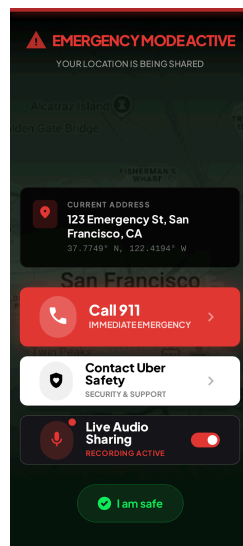
6.Silent alert sent :



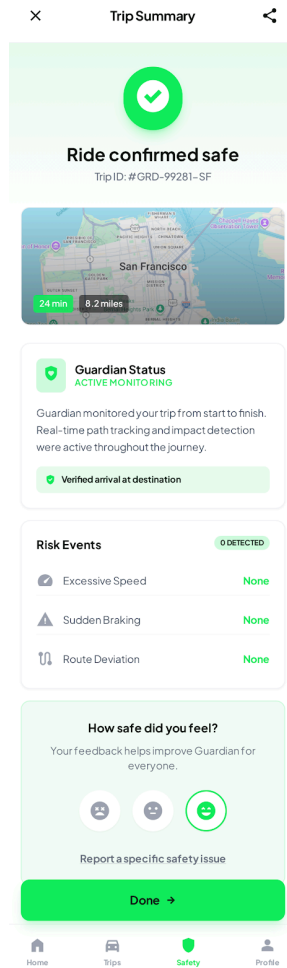
7.Escalated risk state :



8. Emergency mode :



9. Post-Trip summery



8. Edge Cases

1. GPS inaccuracies in tunnels
2. Legitimate route deviation due to traffic
3. Poor network connectivity
4. False positives triggering rider anxiety
5. Driver disputes over anomaly detection
6. Battery saver mode disabling background monitoring
7. Cross-border travel (geo-risk unavailable)

Mitigation:

- Multi-signal validation before escalation

- AI confidence scoring
- Clear driver communication

9. Success Metrics

North Star Metric:

% of rides completed with Guardian Active + Zero Safety Escalations

Metrics	Frequency	Baseline Value Metric	Target Metrics
Safety Feature Engagement Rate (Guardian interactions per ride)	Weekly	23% (current trip sharing usage)	50%+ engagement
Rider Safety Satisfaction Score (Post-Trip Survey)	Monthly	3.9/5	4.6/5
Safety-Related Support Tickets	Monthly	100% (current baseline index)	-15% reduction
Female Rider Retention Rate	Quarterly	Baseline retention X	+8% increase
Night & Solo Ride Bookings	Monthly	Baseline booking volume	+6% growth
Emergency Button False Alarms	Monthly	Current false alarm rate Y	-20% reduction
Severe Safety Incident Rate per 100K Trips	Quarterly	Current incident rate Z	-10% reduction
Auto Ride Sharing Activation Rate	Weekly	23% (manual trip sharing)	60% (auto-enabled under risk conditions)
Guardian Check-In Response Rate	Weekly	0% (feature does not exist)	75%+ rider response rate
App Store Safety Sentiment Score	Quarterly	Current safety sentiment baseline	+15% improvement

Strategic Positioning

Guardian Mode transforms Uber from a reactive safety platform to a proactive, AI-powered ride protection system — delivering invisible protection with visible reassurance.